



MIDDLE EAST TECHNICAL UNIVERSITY

MMI714

Generative Models for Multimedia

(Dis)Entanglement

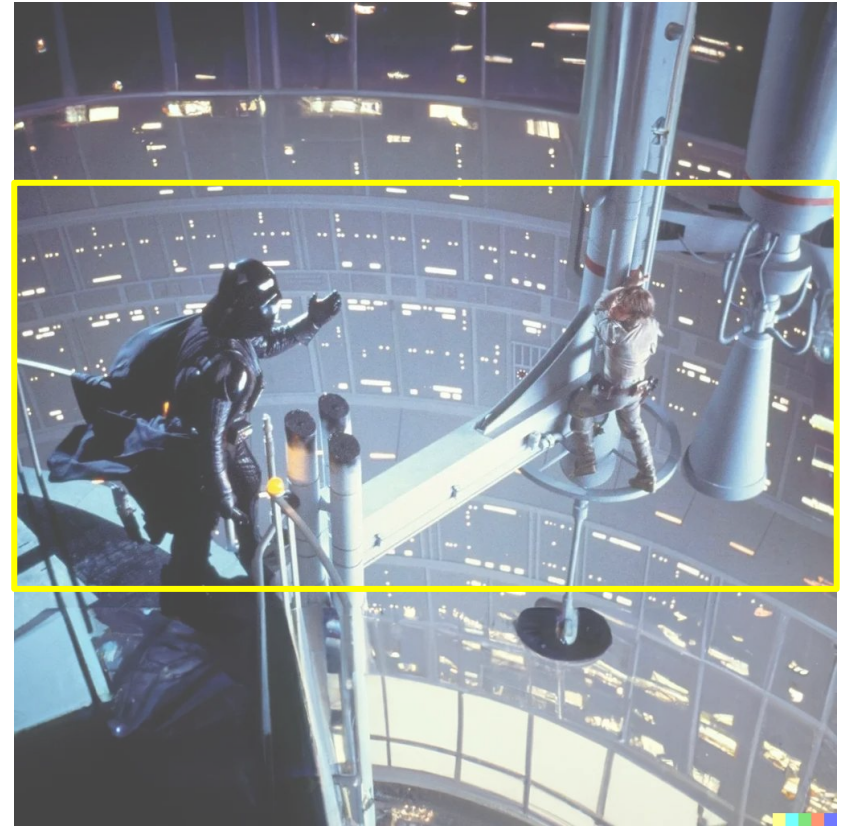
Welcome

This is MMI714

“Generative Models for Multimedia”

This week we will talk about:

Controllability,
(dis)Entanglement



Controllable Generation

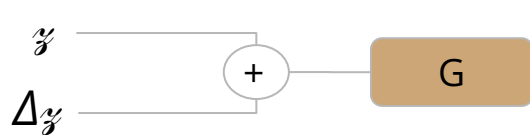
- Controllability vs Conditioning
 - Conditioning involves providing explicit information or constraints during generation, guiding the model to produce output aligned with specified conditions.



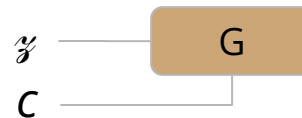
- Controllability emphasizes user-driven customization, allowing manipulation of specific features in the generated output.



Controllable Generation



Controllable



Conditional

Examples with the **features that you want**

Training dataset **doesn't need to be labeled**

Manipulate the z vector
input

Examples from *the classes you want*

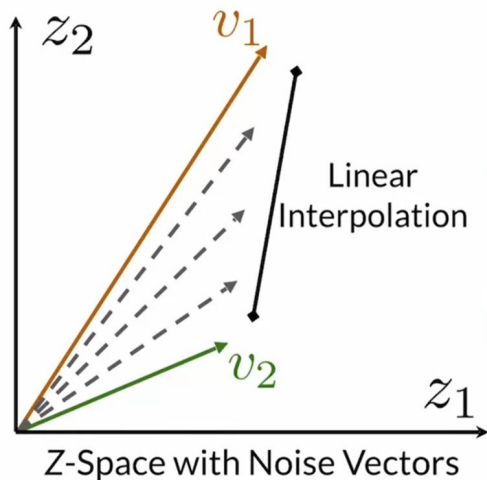
Training dataset *needs to be labeled*

Append a class vector to the
input

Controllable Generation

- Desired case is we can interpolate the z-space with perceptually smooth changes in the target space.

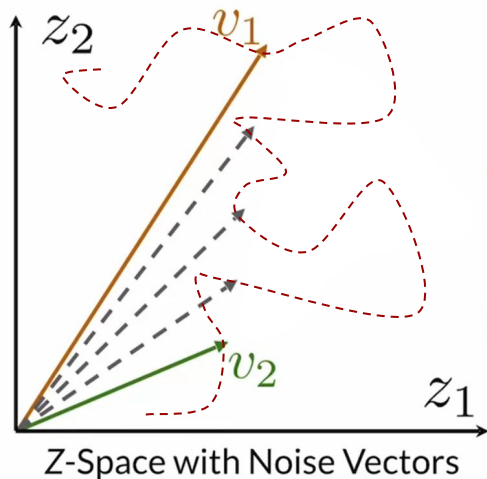
Interpolation Using the Z-Space



Controllable Generation

- However the reality is more like (if you do not specifically design it) this:

Interpolation Using the Z-Space



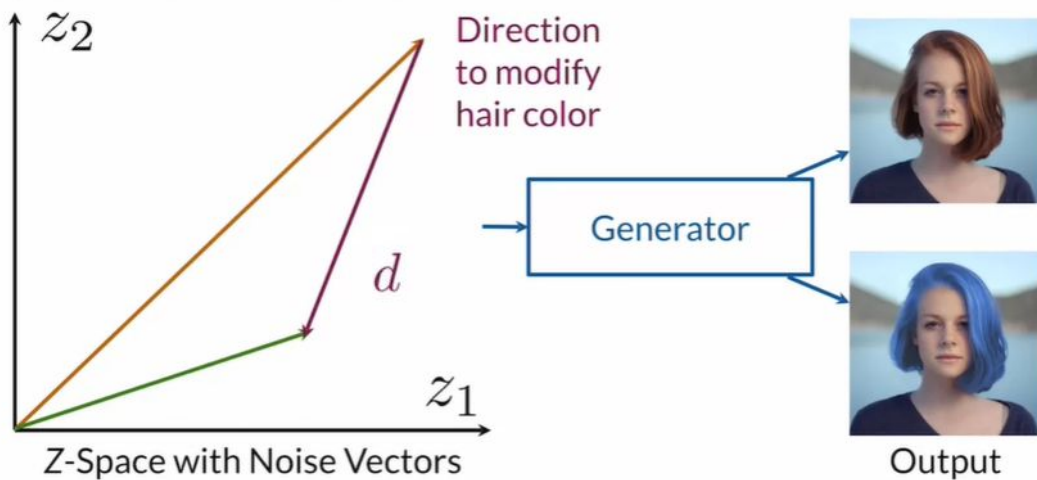
Intermediate images using
the Z-space

$g(v_1)$ $\longleftrightarrow$ $g(v_2)$



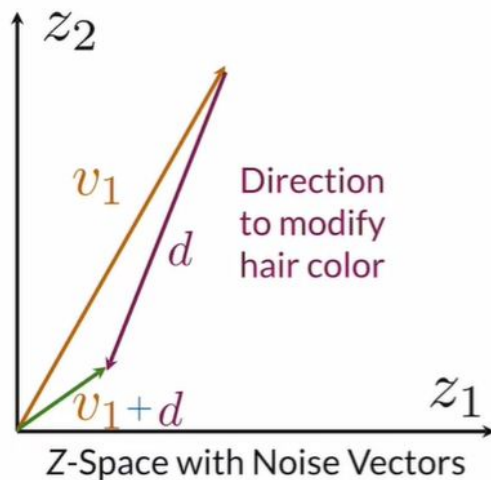
Controllable Generation (desired case)

- In controlling the generation process, our goal is to find “desired” directions.



Controllable Generation (desired case)

- In controlling the generation process, our goal is to find “desired” directions.



Original output

$$g(v_1) \rightarrow$$



Controlled output

$$g(v_1 + d) \rightarrow$$



Only the hair!

Feature Correlation

- The fundamental reason linear interpolation in the latent space is not feasible is because the features are correlated (maybe for the best)



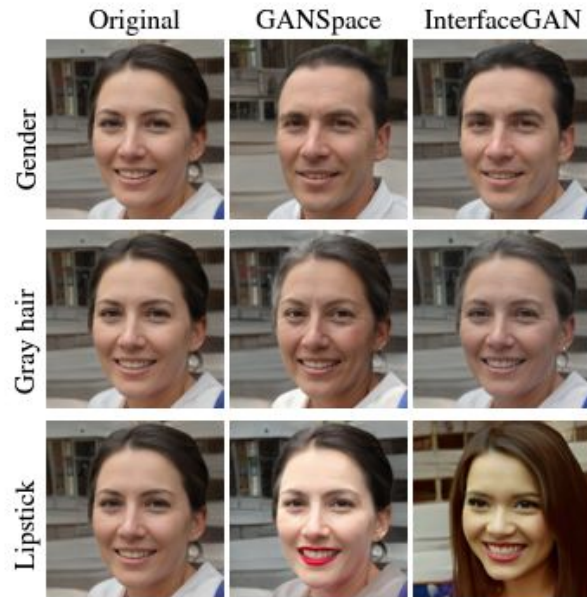
Feature Correlation

- The fundamental reason linear interpolation in the latent space is not feasible is because the features are correlated (maybe for the best)
- What happens is more like this:

Uncorrelated
Features



Correlated
Features

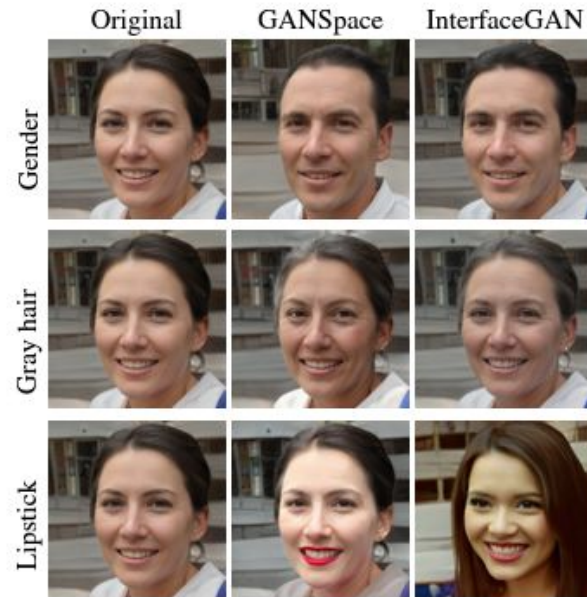


Feature Correlation

- The fundamental reason linear interpolation in the latent space is not feasible is because the features are correlated (maybe for the best)
- What happens is more like this:
- Remember What style GAN had to say about this issue:

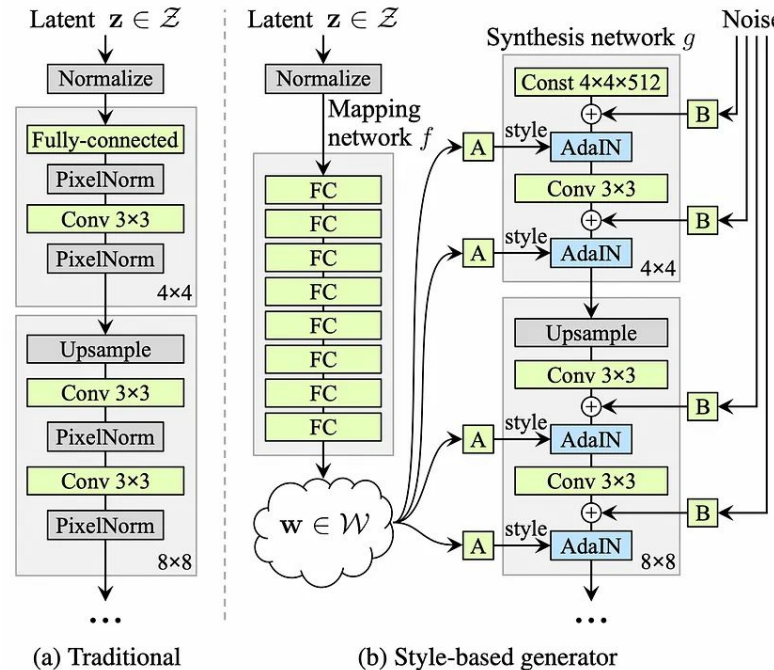
Uncorrelated
Features

Correlated
Features

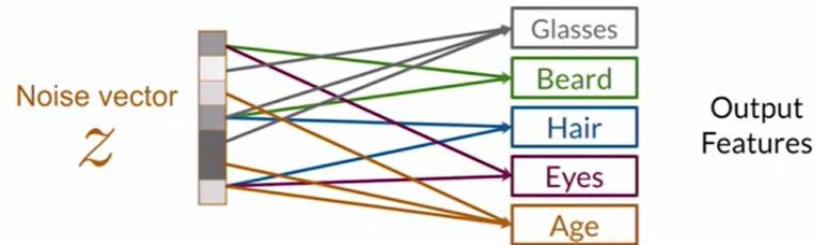


StyleGAN

Our generator embeds the input latent code into an intermediate latent space, which has a profound effect on how the factors of variation are represented in the network. **The input latent space must follow the probability density of the training data, and we argue that this leads to some degree of unavoidable entanglement. Our intermediate latent space is free from that restriction and is therefore allowed to be disentangled.** As previous methods for estimating the degree of latent space disentanglement are not directly applicable in our case, we propose two new automated metrics — perceptual path length and linear separability — for quantifying these aspects of the generator. Using these metrics, we show that compared to a traditional generator architecture, our generator admits a more linear, less entangled representation of different factors of variation.

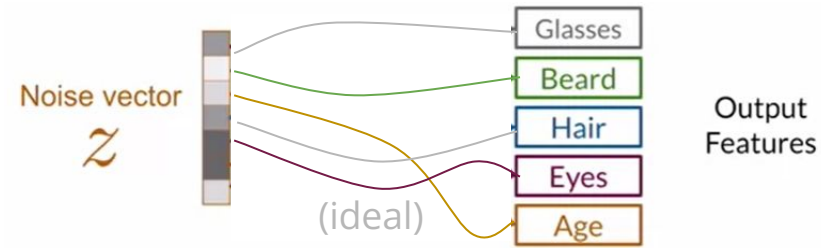


Entanglement



- Entanglement refers to the intricate interdependence of different dimensions or features within the latent space.
- In simpler terms, changes in one dimension impact and influence other dimensions, creating complex relationships.
- The degree of entanglement directly affects the model's capacity to capture and represent diverse and complex patterns in the data.
- Understanding entanglement is crucial for improving the interpretability and controllability of generated outputs.

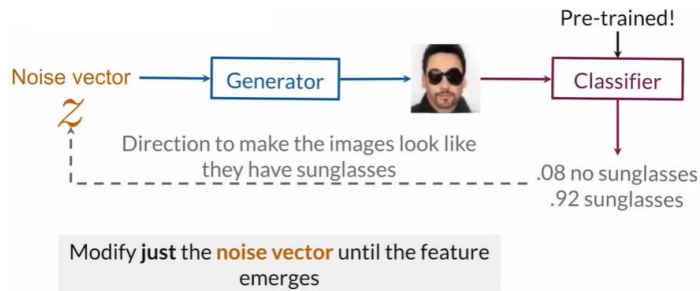
Entanglement



- Entanglement refers to the intricate interdependence of different dimensions or features within the latent space.
- In simpler terms, changes in one dimension impact and influence other dimensions, creating complex relationships.
- The degree of entanglement directly affects the model's capacity to capture and represent diverse and complex patterns in the data.
- Understanding entanglement is crucial for improving the interpretability and controllability of generated outputs.

Direction Gradients

- How to find the direction?
- Or even detect if a (sufficiently) uncorrelated direction even exists?
- One very simple way is to use the discriminator (or a classifier) output (assuming that it outputs something non-binary, or by using some other classifier pre-trained for your purpose)



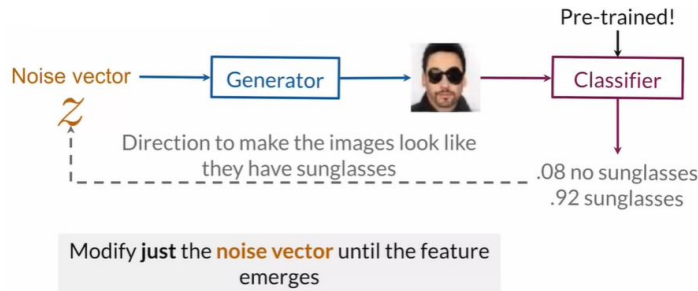
Using a version of SGD with backprop.:

- you can find a $\mathbf{z}$ direction
 - (starting from a given $\mathbf{z}_0$),
- using the partial derivative $\delta y / \delta \mathbf{z}$, and not updating anything else
- where $y = G(D(\mathbf{z}))$

You can even designate a path (possibly a curve) from $\mathbf{z}_0$ to $\mathbf{z}_1$ where you have a higher classifier score.

Direction Gradients

- How to find the direction?
- Or even detect if a (sufficiently) uncorrelated direction even exists?
- One very simple way is to use the discriminator (or a classifier) output (assuming that it outputs something non-binary, or by using some other classifier pre-trained for your purpose)



Using a version of SGD with backprop.:

- you can find a z direction
 - (starting from a given z_0),
- using the partial derivative $\delta y / \delta z$, and not updating anything else
- where $y = G(D(z))$

You can even designate a path (possibly a curve) from z_0 to z_1 where you have a higher classifier score.

DCI Measure(s)

- They address the need for a more objective measure of the quality of learned representations in AI research.
- The goal: “a way to compare models on an equal basis”
- They propose a framework for the quantitative evaluation of disentangled representations

“when the ground-truth latent structure is available”

- **and remove the need for a subjective visual evaluation by a human arbiter (authority).**

A FRAMEWORK FOR THE QUANTITATIVE EVALUATION OF DISENTANGLED REPRESENTATIONS

Cian Eastwood
School of Informatics
University of Edinburgh, UK
c.eastwood@ed.ac.uk

Christopher K. I. Williams
School of Informatics
University of Edinburgh, UK
and Alan Turing Institute, London, UK
ckiw@inf.ed.ac.uk

ABSTRACT

Recent AI research has emphasised the importance of learning *disentangled* representations of the explanatory factors behind data. Despite the growing interest in models which can learn such representations, visual inspection remains the standard evaluation metric. While various desiderata have been implied in recent definitions, it is currently unclear what exactly makes one disentangled representation better than another. In this work we propose a framework for the quantitative evaluation of disentangled representations when the ground-truth latent structure is available. Three criteria are *explicitly* defined and *quantified* to elucidate the quality of learnt representations and thus compare models on an equal basis. To illustrate the appropriateness of the framework, we employ it to compare quantitatively the representations learned by recent state-of-the-art models.

DCI Measure(s)

- Disentanglement - Completeness - Informativeness
- *Disentanglement* measures the degree to which each latent dimension captures at most one attribute,
- *Completeness* measures the degree to which each attribute is controlled by at most one latent dimension,
- *Informativeness* measures the classification accuracy of the attributes, given the latent representation.

A FRAMEWORK FOR THE QUANTITATIVE EVALUATION OF DISENTANGLED REPRESENTATIONS

Cian Eastwood
School of Informatics
University of Edinburgh, UK
c.eastwood@ed.ac.uk

Christopher K. I. Williams
School of Informatics
University of Edinburgh, UK
and Alan Turing Institute, London, UK
ckiw@inf.ed.ac.uk

ABSTRACT

Recent AI research has emphasised the importance of learning *disentangled* representations of the explanatory factors behind data. Despite the growing interest in models which can learn such representations, visual inspection remains the standard evaluation metric. While various desiderata have been implied in recent definitions, it is currently unclear what exactly makes one disentangled representation better than another. In this work we propose a framework for the quantitative evaluation of disentangled representations when the ground-truth latent structure is available. Three criteria are *explicitly* defined and *quantified* to elucidate the quality of learnt representations and thus compare models on an equal basis. To illustrate the appropriateness of the framework, we employ it to compare quantitatively the representations learned by recent state-of-the-art models.

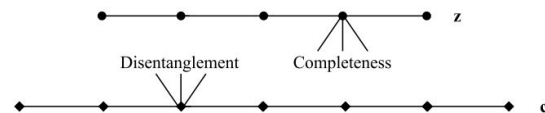


Figure 1: Visualising disentanglement and completeness.

DCI Measure(s)

- Let's follow the paper for the method:

A FRAMEWORK FOR THE QUANTITATIVE EVALUATION OF DISENTANGLED REPRESENTATIONS

Cian Eastwood
School of Informatics
University of Edinburgh, UK
c.eastwood@ed.ac.uk

Christopher K. I. Williams
School of Informatics
University of Edinburgh, UK
and Alan Turing Institute, London, UK
ckiw@inf.ed.ac.uk

ABSTRACT

Recent AI research has emphasised the importance of learning *disentangled* rep-

2 THEORETICAL FRAMEWORK

Models for disentangled factor learning seek a compact data representation or *code* $\mathbf{c}$ of dimension D , which consists of disentangled and interpretable latent variables. For synthetic data, the K -dimensional generative factors $\mathbf{z}$ are designed to be an ideal such representation. Thus if $D = K$ the ideal disentangled code $\mathbf{c}^*$ should be some (scaled) permutation of $\mathbf{z}$, i.e. they should be related by a generalised permutation matrix (or monomial matrix¹). If $D > K$, one would expect to obtain this monomial structure along with a number of ‘dead’ or irrelevant units in $\mathbf{c}$ which are not predictive of / informative about $\mathbf{z}$. Thus, we can quantitatively evaluate the codes learned by a given model M using the following steps:

1. Train M on a synthetic dataset with generative factors $\mathbf{z}$
2. Retrieve $\mathbf{c}$ for each sample $\mathbf{x}$ in the dataset ($\mathbf{c} = M(\mathbf{x})$)
3. Train regressor f to predict $\mathbf{z}$ given $\mathbf{c}$ ($\hat{\mathbf{z}} = f(\mathbf{c})$)
4. Quantify f 's deviation from the ideal mapping and the prediction error

DCI Measure(s)

- In this context
 - c is the learned latent space to be evaluated
 - z is a theoretical/(human designed maybe) “generative factor”
- They choose c a 10 dimensional.
- They choose z as 5 dimensional: $z_0, \dots, z_4$ represent azimuth, elevation, red, green and blue generative factors respectively.

A FRAMEWORK FOR THE QUANTITATIVE EVALUATION OF DISENTANGLED REPRESENTATIONS

Cian Eastwood
School of Informatics
University of Edinburgh, UK
c.eastwood@ed.ac.uk

Christopher K. I. Williams
School of Informatics
University of Edinburgh, UK
and Alan Turing Institute, London, UK
ckiw@inf.ed.ac.uk

ABSTRACT

Recent AI research has emphasised the importance of learning *disentangled* rep-



Figure 2: **Data samples.**

DCI Measure(s)

- In this context
 - c is the learned latent space to be evaluated
 - z is a theoretical/(human designed maybe) “generative factor”
- They choose c a 10 dimensional.
- They choose z as 5 dimensional: $z_0, \dots, z_4$ represent azimuth, elevation, red, green and blue generative factors respectively.

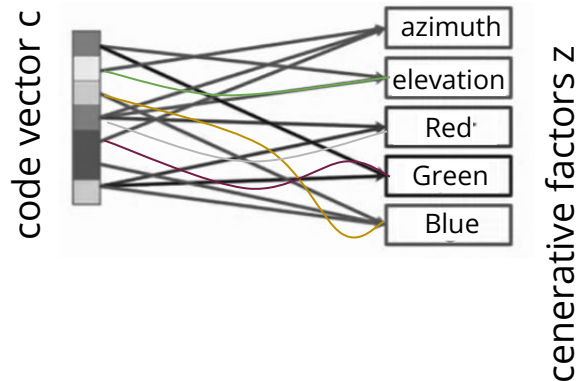


Figure 2: **Data samples.**

DCI Measure(s)

- In this context
 - c is the learned latent space to be evaluated
 - z is a theoretical/(human designed maybe) “generative factor”
- They choose c a 10 dimensional.
- They choose z as 5 dimensional: $z_0, \dots, z_4$ represent azimuth, elevation, red, green and blue generative factors respectively.

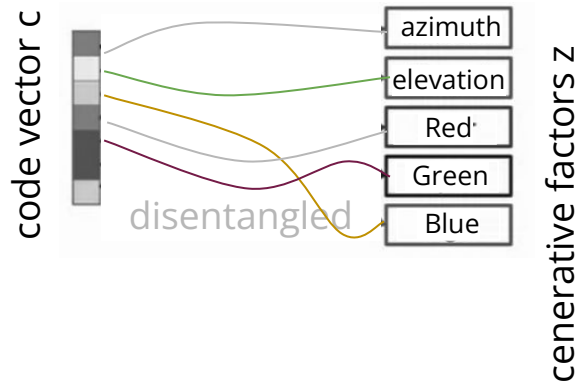


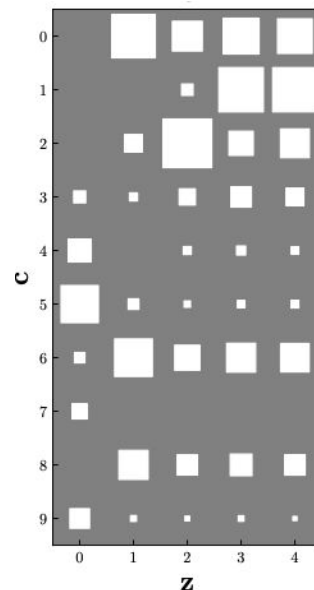
Figure 2: **Data samples.**

Disentanglement

We now detail the proposed evaluation metrics, i.e., steps 3 and 4. We train K regressors to predict the value of K generative factors. The regressor f_j predicts z_j given c , that is, it learns a mapping $f_j(c) : \mathbb{R}^D \rightarrow \mathbb{R}^1$. We use regressors that can provide a matrix of relative importances R , where R_{ij} denotes the relative importance of c_i in predicting z_j (see section 4.3). This allows us to explicitly define and quantify three criteria of disentangled representations or *codes* which are implicit in recent definitions (Desjardins et al., 2012; Bengio et al., 2013; Kulkarni et al., 2015; Chen et al., 2016; Higgins et al., 2017), namely *disentanglement*, *completeness* and *informativeness*.

Disentanglement. The degree to which a representation factorises or *disentangles* the underlying factors of variation, with each variable (or dimension) capturing at most one generative factor. The disentanglement score D_i of code variable c_i is quantified by $D_i = (1 - H_K(P_{i.}))$, where $H_K(P_{i.}) = -\sum_{k=0}^{K-1} P_{ik} \log_K P_{ik}$ denotes the entropy and $P_{ij} = R_{ij} / \sum_{k=0}^{K-1} R_{ik}$ denotes the ‘probability’ of c_i being important for predicting z_j . If c_i is important for predicting a single generative factor, the score will be 1. If c_i is equally important for predicting all generative factors, the score will be 0. D_i can be visualised by examining row i of the Hinton diagrams as in Figure 3.

In order to account for dead or irrelevant units in c , relative code variable importance $\rho_i = \sum_j R_{ij} / \sum_{ij} R_{ij}$ is used to construct a *weighted* average $\sum_i \rho_i D_i$ expressing overall disentanglement. If a code variable c_i is irrelevant for predicting z , then its ρ_i (and thus contribution to the overall disentanglement) will be near zero.



Completeness

A FRAMEWORK FOR THE QUANTITATIVE EVALUATION OF DISENTANGLED REPRESENTATIONS

Completeness. The degree to which each underlying factor is captured by a single code variable. The completeness score C_j in capturing generative factor z_j is quantified by $C_j = (1 - H_D(\tilde{P}_{\cdot j}))$, where $H_D(\tilde{P}_{\cdot j}) = -\sum_{d=0}^{D-1} \tilde{P}_{dj} \log_D \tilde{P}_{dj}$ denotes the entropy of the $\tilde{P}_{\cdot j}$ distribution. If a single code variable contributes to z_j 's prediction, the score will be 1 (complete). If all code variables equally contribute to z_j 's prediction, the score will be 0 (maximally overcomplete). C_j can be visualised by examining column j of the Hinton diagrams as in Figure 3.

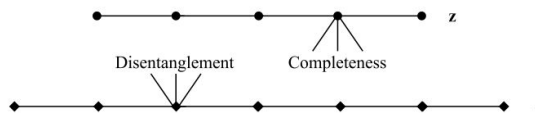
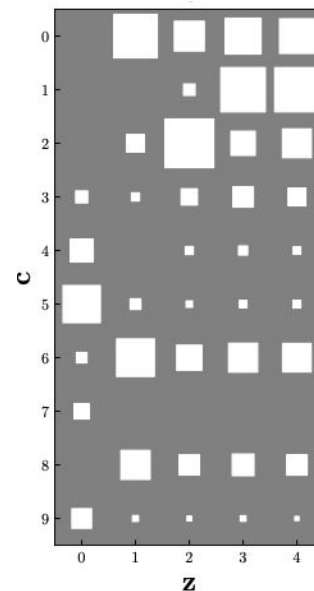


Figure 1: Visualising disentanglement and completeness.

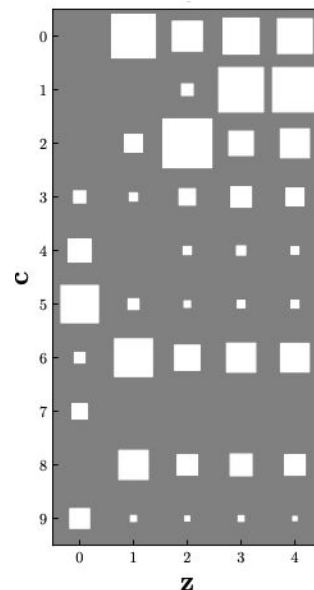
While the disentanglement score quantifies the number of generative factors captured by a given code variable, the completeness score quantifies the number of code variables which capture a given generative factor. Together, these scores quantify the deviation from the ideal one-to-one mapping between z and K of the dimensions in c . Figure 1 illustrates this idea.



Informativeness

A FRAMEWORK FOR THE QUANTITATIVE
EVALUATION OF DISENTANGLED REPRESENTATIONS

Informativeness. The amount of information that a representation captures about the underlying factors of variation. To be useful for natural tasks which require knowledge of the important attributes of the data (e.g. object recognition), representations must ultimately capture information about the underlying factors of variation (Bengio et al., 2013; Chen et al., 2016). The informativeness of code c about generative factor z_j is quantified by the prediction error $E(z_j, \hat{z}_j)$ (averaged over the dataset), where E is an appropriate error function and $\hat{z}_j = f_j(c)$. It is important to note that the prediction error $E(z_j, \hat{z}_j)$, and thus this informativeness metric, is dependent on the capacity of f , with linear regressors only capable of extracting information about z in c that is explicitly represented. Hence this informativeness metric is also dependent on a model's ability to explicitly represent information about z in c , which in turn is dependent on the model's ability to disentangle the underlying factors of variation (z). Thus the informativeness metric has some overlap with the disentanglement metric, with the size of the overlap determined by the capacity of f (no overlap with infinite capacity).



OK. Code Please?

- In this context, code is some learned representations that represent the underlying data representations
 - Like an output of an encoder?
 - YES
- What if there is no encoder like architecture in my model?
 - Well, usually there is.

OK. Code Please?

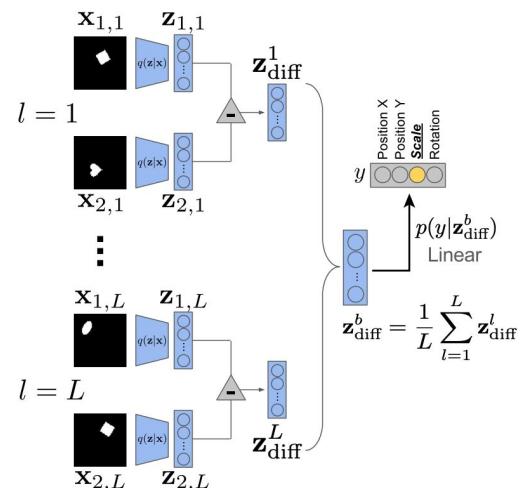
- In this context, code is some learned representations that represent Generative modelling has become one of the leading approaches to unsupervised representation learning, with several recent works imposing additional learning constraints to encourage the model to learn disentangled representations (Desjardins et al., 2012; Reed et al., 2014; Zhu et al., 2014; Cohen & Welling, 2014a; Cheung et al., 2014; Larsen et al., 2015; Makhzani et al., 2015; Chen et al., 2016; Higgins et al., 2017). Of these models, it can be argued that β -VAE (Higgins et al., 2017) and InfoGAN (Chen et al., 2016) are the most promising due to their scalability and lack of assumptions about the underlying factors of variation. Thus, we evaluate the codes learned by these models and compare them to the VAE($\beta = 1$) and PCA.
- All are valid choices, and would represent some level of D, C or I.

β -VAE

VAE vs β -VAE

- The regularization term in beta-VAE is multiplied by a factor $\mathcal{F}(\theta, \phi, \beta; \mathbf{x}, \mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p_{\theta}(\mathbf{x}|\mathbf{z})] - \beta (D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) - \epsilon)$ (3)
- ...and an additional term that penalizes the correlation between the latent variables.
- β -VAE also uses a linear classifier to learn a target factor on the latent variables.

Figure 5: Schematic of the proposed disentanglement metric: over a batch of L samples, each pair of images has a fixed value for one target generative factor y (here $y = scale$) and differs on all others. A linear classifier is then trained to identify the target factor using the average pairwise difference $\mathbf{z}_{diff}^b$ in the latent space over L samples.



<https://openreview.net/pdf?id=Sy2fzU9gl>

Feature	Vanilla VAE	Beta-VAE
Loss function	Reconstruction loss + KL divergence	Reconstruction loss + KL divergence + correlation penalty
Difficulty of training	Easier	More difficult
Sensitivity to hyperparameters	Less sensitive	More sensitive

β -VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK

Irina Higgins, Loic Matthey, Arka Pal, Christopher Burgess, Xavier Glorot, Matthew Botvinick, Shakir Mohamed, Alexander Lerchner
Google DeepMind
{irinah, lmatthey, arkap, cpburgess, glorotx, botvinick, shakir, lerchner}@google.com

ABSTRACT

Learning an interpretable factorised representation of the independent data generative factors of the world without supervision is an important precursor for the development of artificial intelligence that is able to learn and reason in the same way that humans do. We introduce β -VAE, a new state-of-the-art framework for automated discovery of interpretable factorised latent representations from raw image data in a completely unsupervised manner. Our approach is a modification of the variational autoencoder (VAE) framework. We introduce an adjustable hyperparameter β that balances latent channel capacity and independence constraints with reconstruction accuracy. We demonstrate that β -VAE with appropriately tuned $\beta > 1$ qualitatively outperforms VAE ($\beta = 1$), as well as state of the art unsupervised (InfoGAN) and semi-supervised (DC-IGN) approaches to disentangled factor learning on a variety of datasets (*celebA*, *faces* and *chairs*). Furthermore, we devise a protocol to quantitatively compare the degree of disentanglement learnt by different models, and show that our approach also significantly outperforms all baselines quantitatively. Unlike InfoGAN, β -VAE is stable to train, makes few assumptions about the data and relies on tuning a single hyperparameter β , which can be directly optimised through a hyperparameter search using weakly labelled data or through heuristic visual inspection for purely unsupervised data.

InfoGAN (in a nutshell)

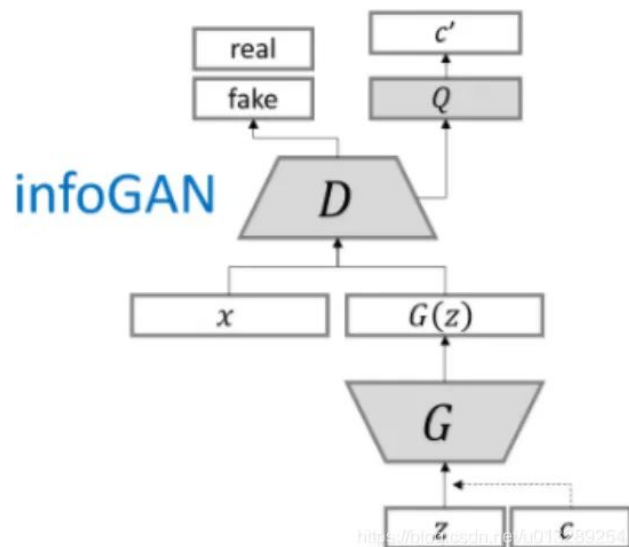
- InfoGAN is *an information-theoretic extension to the GANS that is able to learn disentangled representations in a completely unsupervised manner.*
- It modifies the the GAN loss to encourage it to learn interpretable and meaningful representations.

$$\min_{G,Q} \max_D V_{\text{INFOGAN}}(D, G, Q) = V(D, G) - \lambda L_I(G, Q)$$

Where Q is an auxiliary distribution that approximates the posterior $P(c | x)$ - the probability of the latent code c given the data x - and L_I is the variational lower bound of the mutual information between the latent code and the observations.

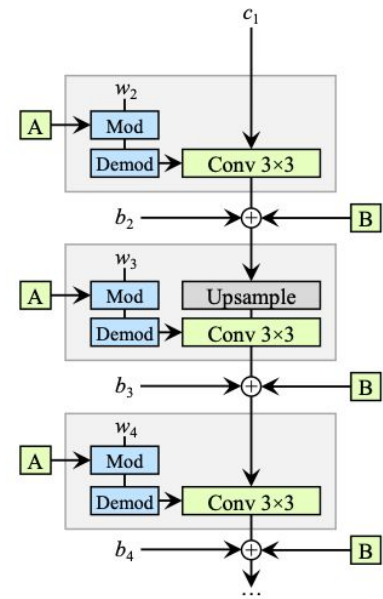
Abstract

This paper describes InfoGAN, an information-theoretic extension to the Generative Adversarial Network that is able to learn disentangled representations in a completely unsupervised manner. InfoGAN is a generative adversarial network that also maximizes the mutual information between a small subset of the latent variables and the observation. We derive a lower bound of the mutual information objective that can be optimized efficiently. Specifically, InfoGAN successfully disentangles writing styles from digit shapes on the MNIST dataset, pose from lighting of 3D rendered images, and background digits from the central digit on the SVHN dataset. It also discovers visual concepts that include hair styles, presence/absence of eyeglasses, and emotions on the CelebA face dataset. Experiments show that InfoGAN learns interpretable representations that are competitive with representations learned by existing supervised methods.



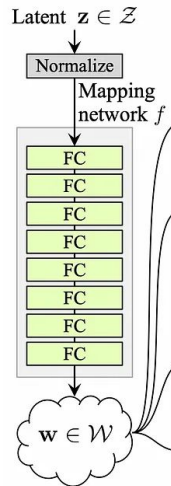
StyleGANs

- Remember StyleGAN2
- Z is transformed into an intermediate latent space, W ,
- Each point in W is further transformed into channel-wise style parameters using a learned affine transformation for each layer of the generator.
- The space spanned by these style parameters is referred to as the StyleSpace (S).
- In a 1024x1024 StyleGAN2 with 18 layers, the StyleSpace (S) has a total of 9088 dimensions



Style Space

- The Style Space is an ideal hyperspace of the channel-wise style parameters, which is supposedly more disentangled.
 - w space was found to better reflect the disentangled nature of the learned distribution with the inception of StyleGAN.
 - To further control the “style”, w can further be transformed to channel-wise style parameters $s \in \mathbf{S}$.
(or some call $w+$ [Abdal et al., 2019])
 - Let’s call the new space $\mathbf{S}$ (or $w+$) : the “style space”
 - $\mathbf{S}$ is (with no surprise) is a relatively higher-dimensional space.
- How to obtain it?





Style Space

Zongze Wu
Hebrew University
zongze.wu@mail.huji.ac.il

Dani Lischinski
Hebrew University
danix@cs.huji.ac.il

Eli Shechtman
Adobe Research
elish@adobe.com

Abstract

We explore and analyze the latent style space of StyleGAN2, a state-of-the-art architecture for image generation, using models pretrained on several different datasets. We first show that StyleSpace, the space of channel-wise style parameters, is significantly more disentangled than the other intermediate latent spaces explored by previous works. Next, we describe a method for discovering a large collection of style channels, each of which is shown to control a distinct visual attribute in a highly localized and disentangled manner. Third, we propose a simple method for identifying style channels that control a specific attribute, using a pretrained classifier or a small number of example images. Manipulation of visual attributes via these StyleSpace controls is shown to be better disentangled than via those proposed in previous works. To show this, we make use of a newly proposed Attribute Dependency metric. Finally, we demonstrate the applicability of StyleSpace controls to the manipulation of real images. Our findings pave the way to semantically meaningful and well-disentangled

wise activation statistics at each of the generator's convolution layers. BigGAN [6] uses class-conditional BatchNorm [14], while StyleGAN [17] uses AdaIN [13] to modulate channel-wise means and variances. StyleGAN2 [18] controls channel-wise variances by modulating the weights of the convolution kernels. It has been shown that the intermediate latent space is more disentangled than the initial one [17]. Additionally, Shen *et al.* [30] show that the latent space of StyleGAN [17, 18] is more disentangled than that of Progressive GAN [16].

Some control over the generated results may be obtained via conditioning [21], which requires training the model with annotated data. In contrast, style-based design enables discovering a variety of interpretable generator controls after training the generator. However, current methods require either a pretrained classifier [10, 30, 31, 35], a large set of paired examples [15], or manual examination of many candidate control directions [12], which limits the versatility of these approaches. Furthermore, the individual controls discovered by these methods are typically entangled, affecting multiple attributes, and are often non-local.

- The Style Space is an ideal hyperspace of parameters, which is supposedly more disentangled than the other intermediate latent spaces explored by previous works.
 - w space was found to better reflect the structure of the learned distribution with the intermediate latent spaces
 - To further control the “style”, w can be mapped to channel-wise style parameters $s \in \mathbf{S}$. (or some call w^+ [Abdal et al., 2019])
 - Let’s call the new space $\mathbf{S}$ (or w^+) : the “style space”
 - $\mathbf{S}$ is (with no surprise) is a relatively higher-dimensional space.
- How to obtain it?

Style Space

- Their goal is to understand
 - to what degree disentanglement is inherent in style-based generator architectures.
 - How to find these disentangled controls?
 - Can this be done in an unsupervised manner, or with only a small amount of supervision?

Zongze Wu

Hebrew University

zongze.wu@mail.huji.ac.il

Dani Lischinski

Hebrew University

danix@cs.huji.ac.il

Eli Shechtman

Adobe Research

elish@adobe.com

Abstract

We explore and analyze the latent style space of StyleGAN2, a state-of-the-art architecture for image generation, using models pretrained on several different datasets. We first show that StyleSpace, the space of channel-wise style parameters, is significantly more disentangled than the other intermediate latent spaces explored by previous works. Next, we describe a method for discovering a large collection of style channels, each of which is shown to control a distinct visual attribute in a highly localized and disentangled manner. Third, we propose a simple method for identifying style channels that control a specific attribute, using a pretrained classifier or a small number of example images. Manipulation of visual attributes via these StyleSpace controls is shown to be better disentangled than via those proposed in previous works. To show this, we make use of a newly proposed Attribute Dependency metric. Finally, we demonstrate the applicability of StyleSpace controls to the manipulation of real images. Our findings pave the way to semantically meaningful and well-disentangled

wise activation statistics at each of the generator's convolution layers. BigGAN [6] uses class-conditional BatchNorm [14], while StyleGAN [17] uses AdaIN [13] to modulate channel-wise means and variances. StyleGAN2 [18] controls channel-wise variances by modulating the weights of the convolution kernels. It has been shown that the intermediate latent space is more disentangled than the initial one [17]. Additionally, Shen *et al.* [30] show that the latent space of StyleGAN [17, 18] is more disentangled than that of Progressive GAN [16].

Some control over the generated results may be obtained via conditioning [21], which requires training the model with annotated data. In contrast, style-based design enables discovering a variety of interpretable generator controls after training the generator. However, current methods require either a pretrained classifier [10, 30, 31, 35], a large set of paired examples [15], or manual examination of many candidate control directions [12], which limits the versatility of these approaches. Furthermore, the individual controls discovered by these methods are typically entangled, affecting multiple attributes, and are often non-local.

Attribute Space

- is huge!

Abstract

We explore and analyze the latent style space of StyleGAN2, a state-of-the-art architecture for image generation, using models pretrained on several different datasets. We first show that StyleSpace, the space of channel-wise style parameters, is significantly more disentangled than the other intermediate latent spaces explored by previous works. Next, we describe a method for discovering a large collection of style channels, each of which is shown to con-

wise activation statistics at each of the generator's convolution layers. BigGAN [6] uses class-conditional BatchNorm [14], while StyleGAN [17] uses AdaIN [13] to modulate channel-wise means and variances. StyleGAN2 [18] controls channel-wise variances by modulating the weights of the convolution kernels. It has been shown that the intermediate latent space is more disentangled than the initial one [17]. Additionally, Shen *et al.* [30] show that the latent space of StyleGAN [17, 18] is more disentangled than that of Progressive GAN [16].

Some control over the generated results may be obtained via conditioning [21], which requires training the model with annotated data. In contrast, style-based design enables discovering a variety of interpretable generator controls after training the generator. However, current methods require either a pretrained classifier [10, 30, 31, 35], a large set of paired examples [15], or manual examination of many candidate control directions [12], which limits the versatility of these approaches. Furthermore, the individual controls discovered by these methods are typically entangled, affecting multiple attributes, and are often non-local.

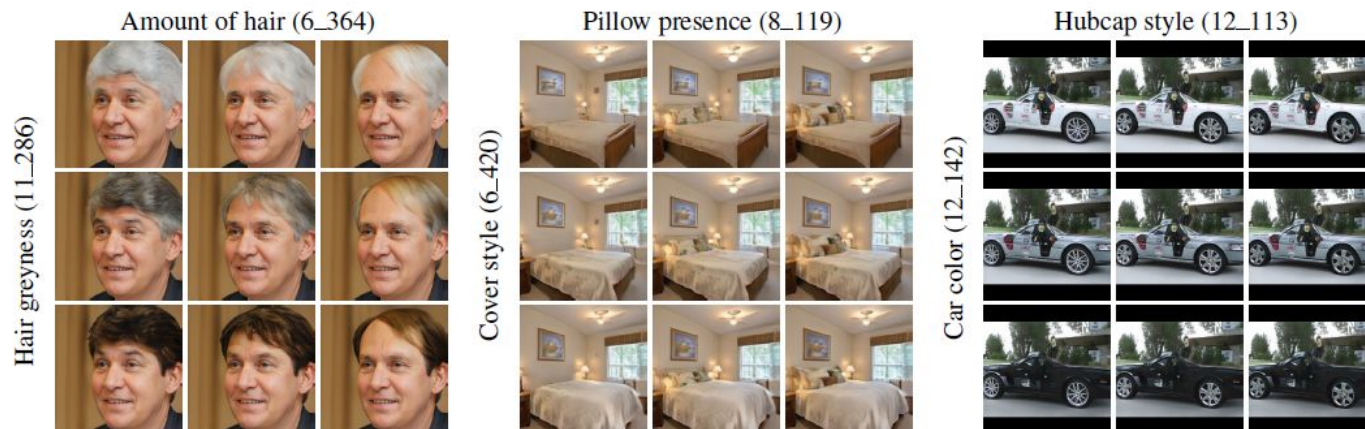


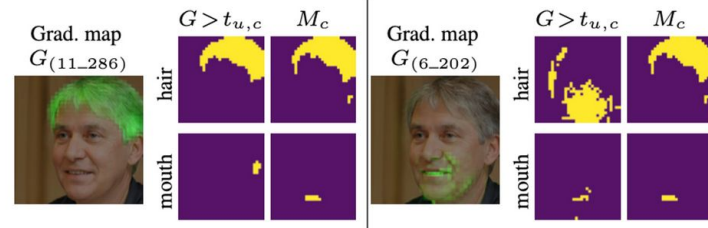
Figure 1. Disentanglement in style space, demonstrated using three different datasets. Each of the three groups above shows two mutations that occur independently inside the same semantic region (hair, bed, and car, from left to right). The indices of the manipulated layer and channel are indicated in parentheses.

Style Space

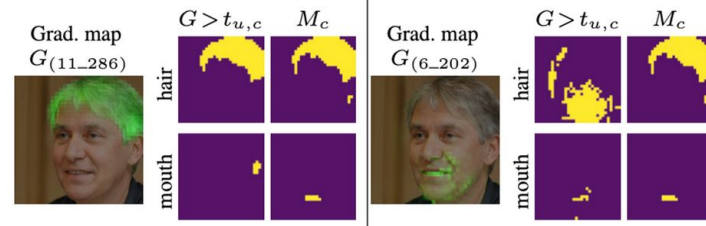
- should also be huge!
- w^+ in [Abdal et al., 2019] converts w (512) into w^+ (9216)
- $\mathbf{S}$ in [Wu et al., 2021] converts w (512) into $\mathbf{s}$ (9088)
- For both papers, the goal is to determine if a latent space offers the a disentangled representation.
- They need a measure for this and use the DCI measure
 - DCI: (disentanglement / completeness/ informativeness)

Style Channels

- To determine the channels that control specific local semantic regions,
- they examine the gradient maps of generated images w.r.t different channels, measure the overlap with specific semantic regions,
- and then with this information, they can identify those changes that are consistently active in each region.



Style Channels



- For each generated image, they apply back-propagation to compute the gradient map of the image with respect to each channel of the StyleSpace.
- This gradient map represents how changes in the values of the style channels affect the generated image.
- By examining these gradient maps and measuring their overlap with specific semantic regions, they can identify those channels that are consistently active in each region, indicating that they control the appearance of those local semantic regions

Style Channels

Figure 3. Examples of manipulations, each controlled by a single style channel. Each pair of images shows the result of manipulation by decreasing (-) and increasing (+) the value of the style parameter (the original image is omitted). The layer index, channel index, and the direction of change is overlaid in the bottom left corner.

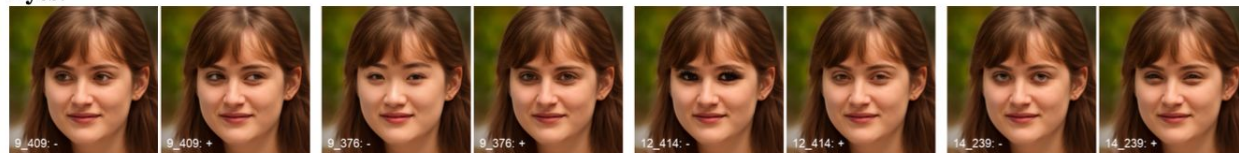
Hair:



Mouth:



Eyes:



Style on real data

- To manipulate real images, there are two ways latent optimization or by training an encoder
- The latent optimization (from Style GAN 2 paper) process involves iteratively adjusting the latent code to minimize the difference between the generated image and the target real image.
- This is typically done by optimizing a loss function, such as the L2 loss, which measures the discrepancy between the generated image and the target image.

Style Channels

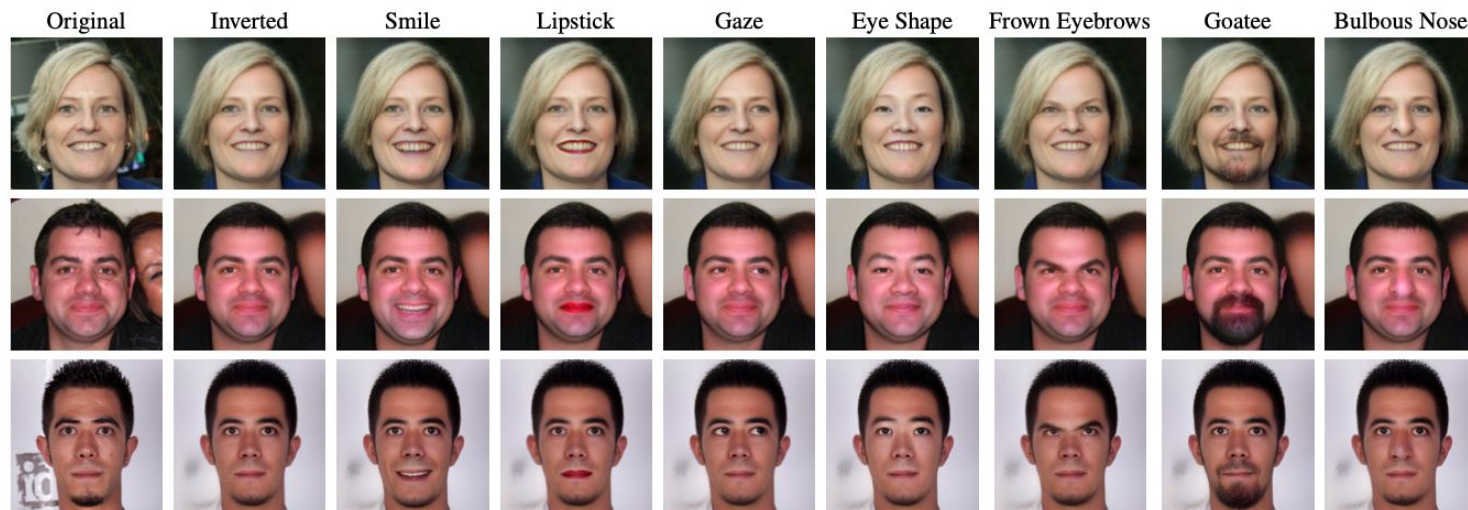


Figure 7. Manipulation of real images using encoder-based inversion. Original images are from FFHQ, and were not part of the encoder's training set. More results can be found in app. Figure [20](#).

Thank you!



“an artistic painting of generalized artificial intelligence. AI is symbolized with a child robot, holding his human parent's hand.”

 Erdem x DALL-E
Human & AI